

Related Work Table

Title	Authors	Years	Dataset	Application area	Method	Accuracy	Limitations
Interpretable Deep Learning Framework for LULC Classification in Remote Sensing Using SHAP	Temenos et al.	2023	Interpretable Deep Learning Framework for LULC Classification in Remote Sensing Using SHAP	Land use/land cover classification with band-importance analysis	Compact CNN + SHAP (deep explainer)	94.72%	Accuracy capped below other CNN benchmarks (~98%); SHAP explanation is computationally expensive
Kolmogorov-Arnold Network for Satellite Image Classification in Remote Sensing	Cheon, M.	2024	EuroSat --Sentinel-2 satellite images, 10 LULC classes	Satellite/land use image classification, exploring a novel MLP-alternative architecture	Kolmogorov-Arnold Network, tested alongside ConvNext, ResNet101, VGG16, MobileNetV2, EfficientNet, ViT	Best result: ConvNeXt + KAN reaching 96% accuracy (starting at 94% in epoch 1)	Only a preprint accuracy (96%) is below top CNN/ViT benchmarks on EuroSat (~98%+); KAN training is less mature/optimized than standard architectures
Explainable Custom CNN Architecture for Land Use Classification using Satellite Images	Verma, M. et al.	2021	EuroSat - 27,000 geo-referenced Sentinel-2 images, 10 classes	Land use classification with model interpretability	Custom CNN + LIME	88.21%	Lower accuracy than other EuroSat benchmarks, explanations are qualitative, not band-level

Mapping of Land Use and Land Cover (LULC) using EuroSAT and Transfer Learning	Kunwar, S., Ferdush, J.	2023	EuroSat (RGB) - 27,000 images, 64×64 px, 10 classes	LULC mapping applied to a real region (Kreis Borken, Germany)	Vision Transformer (ViT) pre-trained on ImageNet-21k, fine-tuned via transfer learning.	99.19%	Uses only RGB bands transformer training needs GPU resources.
Benchmarking and Scaling of Deep Learning Models for Land Cover Image Classification	Papoutsis et al.	2023	BigEarthNet (Sentinel-2, multi-label) — evaluated further on EuroSat and SEN12MS	Large-scale LULC benchmarking and transfer learning	CNN zoo (60+ models), EfficientNet-scaled Wide ResNet + Efficient Channel Attention	+4.5% F1-score over ResNet50 baseline (BigEarthNet, 19 classes)	EuroSat is a secondary evaluation, not primary focus; large-scale setup is compute-heavy